

Unbounded ratios in Erdős Problem 1054

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Abstract

Let $f(N)$ be the least positive integer m for which N is the sum of an initial segment of the positive divisors of m , arranged in increasing order. We prove that, for every fixed $A \geq 1$, a positive proportion of integers N satisfy both $f(N) < \infty$ and $f(N) > AN$. In particular,

$$\limsup_{\substack{N \rightarrow \infty \\ f(N) < \infty}} \frac{f(N)}{N} = \infty.$$

The proof has two complementary parts. Prefix representations with a bounded cofactor are eliminated on a sifted set of positive density, while a third-moment estimate shows that representations with a large cofactor and $m \leq AN$ occupy a smaller set. The classical theorem that almost every even integer is a sum of two primes is used only to ensure that a density-one subset of the surviving odd integers is represented at all.

1 Statement, conventions, and prior work

For $m \geq 1$, list its positive divisors as

$$1 = d_1(m) < d_2(m) < \cdots < d_{\tau(m)}(m) = m.$$

An integer $N \geq 1$ is called *represented* if

$$N = d_1(m) + \cdots + d_k(m)$$

for some $m \geq 1$ and some $1 \leq k \leq \tau(m)$. Let $\mathcal{R}$ be the set of represented integers and define

$$f(N) := \min\{m \geq 1 : N = d_1(m) + \cdots + d_k(m) \text{ for some } k\}$$

for $N \in \mathcal{R}$. When convenient, we extend the definition by setting $f(N) = \infty$ for $N \notin \mathcal{R}$.

The question was attributed to Erdős and recorded as Problem B2 in Guy's collection [6]; the equivalent formulation in which the divisor 1 is omitted appears in Erdős and Graham [5, p. 93]. The two functions differ only by a shift of the argument. The problem is catalogued online as Erdős Problem 1054, with the shifted version as Problem 468 [2, 3], and its computed values form OEIS sequence A167485 [13]. The OEIS entry also records the elementary observation $f(\sigma(n)) \leq n$, which implies $\liminf_{N \rightarrow \infty} f(N)/N = 0$ along represented integers because $\sigma(n)/n$ is unbounded [13].

The previously known results concern the opposite, small-ratio tail. Tao proved in an online comment that

$$\#\{N \leq X : f(N) \leq \delta N\} \ll \delta^2 X$$

uniformly in X [19]. This disproved both the pointwise and density-one forms of $f(N) = o(N)$. Subsequent online work gave higher-moment variants: Price posted links to AI-generated arguments giving arbitrarily high fixed polynomial decay [16]. Kovač then gave a self-contained

argument, incorporating an optimization suggested by Bloom, proving that for some $c > 0$ and all sufficiently small $\delta > 0$,

$$\sup_{X \geq 1} \frac{1}{X} \# \{N \leq X : f(N) \leq \delta N\} \ll \exp\left(-e^{(1/\delta)^c}\right).$$

See [9, 4]. These estimates do not imply that $f(N)/N$ is unbounded. Kovač explicitly identified the limsup question as a separate remaining problem [10].

The possible nonexistence of $f(N)$ was also discussed in the same forum. jif supplied a computer-assisted argument, together with a verifier, that the represented integers are exactly $\mathbb{N} \setminus \{2, 5\}$ [17, 18]. The proof below does not depend on that computation: it proves directly that almost every odd integer is represented, which is enough for the limsup statement.

Our main result is the following stronger form of the desired conclusion.

Theorem 1.1. *For every fixed real number $A \geq 1$, there exists a constant $c_A > 0$ such that*

$$\liminf_{X \rightarrow \infty} \frac{1}{X} \# \{N \leq X : N \in \mathcal{R}, f(N) > AN\} \geq c_A.$$

Consequently,

$$\limsup_{\substack{N \rightarrow \infty \\ N \in \mathcal{R}}} \frac{f(N)}{N} = \infty.$$

2 Three standard analytic inputs

We use the following classical facts.

(A1) For every prime q ,

$$\sum_{\substack{p \text{ prime} \\ p \equiv -1 \pmod{q}}} \frac{1}{p} = \infty.$$

This is the reciprocal-prime form of Dirichlet's theorem, or equivalently a weak form of Mertens' theorem in arithmetic progressions.

(A2) There is an absolute constant $c_0 > 0$ such that, for all $y \geq 2$,

$$\prod_{p \leq y} \left(1 - \frac{1}{p}\right) \geq \frac{c_0}{\log(2y)}.$$

(A3) Let

$$E_G(X) := \# \{n \leq X : n \text{ is even and is not a sum of two primes}\}.$$

Then $E_G(X) = o(X)$.

Facts (A1) and (A2) are standard consequences of the prime number theorem for arithmetic progressions and Mertens' theorem. Fact (A3) is the classical “almost all” binary Goldbach theorem. Montgomery and Vaughan prove the stronger estimate $E_G(X) \ll X^{1-\delta}$ for some effectively computable constant $\delta > 0$, which suffices here [12, Theorem 1]. We shall also use the elementary consequence $\pi(X) = o(X)$.

3 Exact parametrization of divisor prefixes

The change of variables below is the fixed-cofactor form of the divisor-reflection device used in Tao's comment and in Kovač's higher-moment treatment [19, 9]. We record the exact form needed for the present prefix convention.

For integers $e, d \geq 1$, define

$$F_e(d) := \sum_{\substack{q|ed \\ q \leq d}} q.$$

Thus $F_e(d)$ is the divisor-prefix sum of ed whose final divisor is d . Also define

$$g_e(n) := \sum_{\substack{r|n \\ r \geq e}} \frac{1}{r}.$$

Lemma 3.1 (Reflection identity). *For all $e, d \geq 1$,*

$$F_e(d) = ed g_e(ed).$$

Every divisor-prefix representation of an integer N is of the form

$$N = F_e(d), \quad m = ed,$$

for some $e, d \geq 1$; conversely every such pair gives a divisor-prefix representation.

Proof. Under divisor reflection $r \mapsto ed/r$, the condition $r \geq e$ is equivalent to $ed/r \leq d$. Hence

$$ed g_e(ed) = \sum_{\substack{r|ed \\ r \geq e}} \frac{ed}{r} = \sum_{\substack{q|ed \\ q \leq d}} q = F_e(d).$$

If a prefix of the divisors of m ends at the divisor $d \mid m$, then $e = m/d$ and its sum is $F_e(d)$. The converse is immediate because $d \mid ed$. $\square$

The parameter $e = m/d$ will be called the *cofactor* of the prefix endpoint.

4 A density-zero lemma for the sum-of-divisors function

The principal result in this section is classical. Alaoglu and Erdős observed that, for every fixed prime q , one has $q \mid \sigma(n)$ for a density-one set of integers n [1, p. 882]. See also [15, Theorem 2], [14, Lemma 5], and the modern discussion in [11, §1]. We include the short sieve proof because the precise formulation is used in the bounded-cofactor argument.

Write $p \parallel n$ when $p \mid n$ but $p^2 \nmid n$.

Lemma 4.1. *For every prime q , the set*

$$\mathcal{B}_q := \{n \geq 1 : q \nmid \sigma(n)\}$$

has natural density zero. Equivalently, if

$$B_q(Y) := \#\{n \leq Y : q \nmid \sigma(n)\},$$

then $B_q(Y) = o(Y)$.

Proof. Let

$$\mathcal{P}_q := \{p \text{ prime} : p \equiv -1 \pmod{q}\}.$$

By (A1), $\sum_{p \in \mathcal{P}_q} 1/p$ diverges. If $p \in \mathcal{P}_q$ and $p \parallel n$, then multiplicativity of σ gives

$$q \mid p+1 = \sigma(p) \mid \sigma(n).$$

Therefore $n \in \mathcal{B}_q$ only if $p \nparallel n$ for every $p \in \mathcal{P}_q$.

Let $\mathcal{P}' \subset \mathcal{P}_q$ be finite. By the Chinese remainder theorem, the natural density of the integers for which $p \nparallel n$ for every $p \in \mathcal{P}'$ is

$$\prod_{p \in \mathcal{P}'} \left(1 - \frac{1}{p} + \frac{1}{p^2}\right),$$

because the density of $p \parallel n$ is $1/p - 1/p^2$. Consequently,

$$\bar{d}(\mathcal{B}_q) \leq \prod_{p \in \mathcal{P}'} \left(1 - \frac{1}{p} + \frac{1}{p^2}\right).$$

Since

$$\sum_{p \in \mathcal{P}_q} \left(\frac{1}{p} - \frac{1}{p^2}\right) = \infty,$$

the products on the right tend to zero as $\mathcal{P}'$ increases through finite subsets of $\mathcal{P}_q$. Thus $\bar{d}(\mathcal{B}_q) = 0$, and hence $\mathcal{B}_q$ has natural density zero. $\square$

We will repeatedly use the following elementary summation principle.

Lemma 4.2 (Summation over a thin multiplicative set). *Let $B(Y)$ be a nonnegative counting function satisfying $B(Y) = o(Y)$ and $B(Y) \leq Y$. Let $\mathcal{A} \subset \mathbb{N}$ satisfy*

$$\sum_{a \in \mathcal{A}} \frac{1}{a} < \infty.$$

Then

$$\sum_{\substack{a \in \mathcal{A} \\ a \leq X}} B(X/a) = o(X).$$

Proof. Fix $T \geq 1$. Dividing by X gives

$$\frac{1}{X} \sum_{\substack{a \in \mathcal{A} \\ a \leq X}} B(X/a) \leq \sum_{\substack{a \in \mathcal{A} \\ a \leq T}} \frac{1}{a} \frac{B(X/a)}{X/a} + \sum_{\substack{a \in \mathcal{A} \\ a > T}} \frac{1}{a}.$$

For fixed T , the first sum tends to zero as $X \rightarrow \infty$. The second sum can be made arbitrarily small by taking T large. This proves the claim. $\square$

5 Bounded cofactors miss a positive-density sifted set

Fix an integer $e \geq 2$. Every $d \geq 1$ has a unique factorization

$$d = ab, \quad a := \prod_{p \leq e} p^{v_p(d)},$$

where every prime factor of a is at most e , every prime factor of b is greater than e , and $(b, ea) = 1$.

Lemma 5.1 (Smooth-rough identity). *With the preceding factorization,*

$$F_e(ab) = bF_e(a) + \sigma(ea)(\sigma(b) - b).$$

Equivalently, if

$$H_e(a) := \sigma(ea) - F_e(a),$$

then

$$F_e(ab) = \sigma(ea)\sigma(b) - H_e(a)b. \quad (5.1)$$

Proof. Because $(ea, b) = 1$, every divisor of $ea b$ has a unique form xy with $x \mid ea$ and $y \mid b$.

If $y = b$, then $xy \leq ab$ if and only if $x \leq a$, and these terms contribute $bF_e(a)$.

If $y < b$, put $c = b/y$. Then $c > 1$ and every prime factor of c is greater than e , so $c > e$. For every $x \mid ea$,

$$xy = x \frac{b}{c} \leq ea \frac{b}{c} < ab.$$

Thus all divisors $x \mid ea$ occur, and the terms with $y < b$ contribute

$$\sigma(ea) \sum_{\substack{y \mid b \\ y < b}} y = \sigma(ea)(\sigma(b) - b).$$

This proves the first identity, and the second is a rearrangement. $\square$

For an e -smooth integer a , define its active set

$$\mathcal{R}_e(a) := \{r : 1 \leq r < e, r \mid ea\}.$$

There are only finitely many possible active sets for a fixed e .

Lemma 5.2 (A forced prime divisor). *For every $e \geq 2$ and every e -smooth a ,*

$$\frac{H_e(a)}{a} = e \sum_{r \in \mathcal{R}_e(a)} \frac{1}{r}. \quad (5.2)$$

For every possible active set $\mathcal{R}$, write the rational number on the right in lowest terms as

$$e \sum_{r \in \mathcal{R}} \frac{1}{r} = \frac{A_{e,\mathcal{R}}}{B_{e,\mathcal{R}}}, \quad (A_{e,\mathcal{R}}, B_{e,\mathcal{R}}) = 1.$$

Then $A_{e,\mathcal{R}} \geq 2$. If $q_{e,\mathcal{R}}$ is any prime divisor of $A_{e,\mathcal{R}}$ and $\mathcal{R}_e(a) = \mathcal{R}$, then

$$q_{e,\mathcal{R}} \mid H_e(a).$$

Proof. The quantity $H_e(a)$ is the sum of those divisors $x \mid ea$ that exceed a . Reflecting x to $r = ea/x$ gives

$$H_e(a) = \sum_{\substack{x \mid ea \\ x > a}} x = ea \sum_{\substack{r \mid ea \\ r < e}} \frac{1}{r},$$

which proves (5.2). Since $1 \in \mathcal{R}_e(a)$, the rational number in (5.2) is at least $e \geq 2$, so its reduced numerator is at least 2.

If $\mathcal{R}_e(a) = \mathcal{R}$, then

$$H_e(a) = a \frac{A_{e,\mathcal{R}}}{B_{e,\mathcal{R}}}.$$

The left side is an integer. Since the numerator and denominator are coprime, $B_{e,\mathcal{R}} \mid a$, and therefore

$$H_e(a) = A_{e,\mathcal{R}} \frac{a}{B_{e,\mathcal{R}}}.$$

Every prime divisor of $A_{e,\mathcal{R}}$ consequently divides $H_e(a)$. $\square$

For each pair $(e, \mathcal{R})$, choose once and for all one such prime $q_{e, \mathcal{R}}$. For an integer $E \geq 2$, let

$$S_E := \{2\} \cup \{q_{e, \mathcal{R}} : 2 \leq e \leq E, \mathcal{R} \text{ is a possible active set for } e\}$$

and define the squarefree integer

$$Q_E := \prod_{q \in S_E} q, \quad \delta_E := \frac{\varphi(Q_E)}{Q_E}.$$

Proposition 5.3 (Bounded-cofactor exclusion). *For every fixed $E \geq 2$,*

$$\#\{N \leq X : (N, Q_E) = 1, \quad N = F_e(d) \text{ for some } 1 \leq e \leq E\} = o_E(X).$$

Proof. First let $2 \leq e \leq E$ be fixed. Write $d = ab$ as above and put $\mathcal{R} = \mathcal{R}_e(a)$ and $q = q_{e, \mathcal{R}}$. If

$$N = F_e(ab), \quad (N, Q_E) = 1,$$

then $q \nmid N$. By Lemma 5.2, $q \mid H_e(a)$. Reducing (5.1) modulo q , we see that $q \mid \sigma(b)$ would force $q \mid N$. Therefore

$$q \nmid \sigma(b). \tag{5.3}$$

Also $F_e(d) \geq d$, because d itself occurs in the defining sum. Hence $N \leq X$ implies $d = ab \leq X$.

For a fixed active set $\mathcal{R}$, the number of possible d is therefore at most

$$\sum_{\substack{a \leq X \\ p \mid a \Rightarrow p \leq e}} B_q(X/a).$$

The set of e -smooth integers has convergent reciprocal sum:

$$\sum_{\substack{a \geq 1 \\ p \mid a \Rightarrow p \leq e}} \frac{1}{a} = \prod_{p \leq e} \left(1 - \frac{1}{p}\right)^{-1} < \infty.$$

Lemma 4.1 and Lemma 4.2 show that the displayed count is $o_e(X)$. There are only finitely many active sets, so the same conclusion holds for this fixed e .

It remains to treat $e = 1$. Here

$$F_1(d) = \sigma(d).$$

Since $2 \mid Q_E$, the condition $(F_1(d), Q_E) = 1$ implies that $\sigma(d)$ is odd. The well-known parity criterion for σ says that this occurs if and only if d is a square or twice a square: for every odd prime p , the factor $1 + p + \dots + p^{v_p(d)}$ is odd exactly when $v_p(d)$ is even, while the factor contributed by $p = 2$ is always odd. Thus there are $O(\sqrt{X})$ possible $d \leq X$.

Finally, sum over the finitely many $e \leq E$. $\square$

We next show that the sifted set $(N, Q_E) = 1$ is not too sparse.

Lemma 5.4 (Density of the sifted set). *There is an absolute constant $c_1 > 0$ such that, for every $E \geq 2$,*

$$\delta_E \geq \frac{c_1}{E \log(2E)}.$$

Moreover,

$$\#\{N \leq X : (N, Q_E) = 1\} = \delta_E X + O_E(1).$$

Proof. Put

$$L_{e-1} := \text{lcm}(1, 2, \dots, e-1).$$

For a possible active set $\mathcal{R}$,

$$e \sum_{r \in \mathcal{R}} \frac{1}{r} = \frac{e}{L_{e-1}} \sum_{r \in \mathcal{R}} \frac{L_{e-1}}{r}.$$

The unreduced numerator is at most

$$e(e-1)L_{e-1}.$$

Hence

$$q_{e,\mathcal{R}} \leq A_{e,\mathcal{R}} \leq e(e-1)L_{e-1} \leq E^2 L_{E-1} =: Y_E.$$

The additional prime 2 also satisfies $2 \leq Y_E$. Thus S_E is a subset of the primes not exceeding Y_E , and

$$\delta_E = \prod_{q \in S_E} \left(1 - \frac{1}{q}\right) \geq \prod_{p \leq Y_E} \left(1 - \frac{1}{p}\right).$$

By (A2), the last product is $\gg 1/\log(2Y_E)$. Since

$$L_{E-1} \leq (E-1)! \leq E^{E-1},$$

we have $Y_E \leq E^{E+1}$ and therefore

$$\log(2Y_E) \ll E \log(2E).$$

This proves the lower bound for δ_E .

The counting formula follows from periodicity modulo the fixed integer Q_E : in each complete residue block of length Q_E , exactly $\varphi(Q_E)$ integers are coprime to Q_E . $\square$

6 A third-moment estimate for large cofactors

The expansion in Lemma 6.1 is the fixed-threshold, third-moment form of the divisor-reflection and least-common-multiple method used by Tao and Kovač [19, 9]. Multiple reciprocal sums involving greatest common divisors and least common multiples have also been studied systematically in [7, 8]. Those works do not directly supply the one-sided threshold estimate in Lemma 6.2, for which we give a separate elementary proof.

For $e \geq 1$, define

$$T_3(e) := \sum_{r,s,t \geq e} \frac{1}{rst [r, s, t]},$$

where $[r, s, t]$ denotes the least common multiple.

Lemma 6.1 (Third moment). *For every $Z \geq 1$ and $e \geq 1$,*

$$\sum_{n \leq Z} g_e(n)^3 \leq Z T_3(e).$$

Proof. Expanding the cube and counting common multiples gives

$$\begin{aligned} \sum_{n \leq Z} g_e(n)^3 &= \sum_{r,s,t \geq e} \frac{1}{rst} \#\{n \leq Z : r \mid n, s \mid n, t \mid n\} \\ &= \sum_{r,s,t \geq e} \frac{1}{rst} \left\lfloor \frac{Z}{[r, s, t]} \right\rfloor \\ &\leq Z T_3(e). \end{aligned}$$

$\square$

Lemma 6.2 (Power-saving tail). *There is an absolute constant $C_3 > 0$ such that*

$$T_3(e) \leq C_3 e^{-5/2} \quad (e \geq 1).$$

Consequently, for another absolute constant $C_4 > 0$,

$$\sum_{e > E} T_3(e) \leq C_4 E^{-3/2} \quad (E \geq 1).$$

Proof. We use a prime-by-prime decomposition of a triple (r, s, t) . For every prime p , let the three valuations $v_p(r), v_p(s), v_p(t)$ have minimum λ_0 , median λ_1 , and maximum λ_2 . Put p^{λ_0} into a variable h . Put $p^{\lambda_1 - \lambda_0}$ into one of u, v, w according as the two variables whose valuations are at least λ_1 are (r, s) , (r, t) , or (s, t) . Finally, put $p^{\lambda_2 - \lambda_1}$ into one of R, S, T according as the variable with largest valuation is r, s, t , respectively. If an exponent difference is zero, the corresponding choice is immaterial.

This gives positive integers h, u, v, w, R, S, T such that

$$r = huvR, \quad s = huwS, \quad t = hvwT,$$

and, by construction,

$$rst = h^3 u^2 v^2 w^2 RST, \quad [r, s, t] = huvwRST. \quad (6.1)$$

Every triple (r, s, t) determines such a seven-tuple. We shall bound the sum over the resulting canonical tuples by the larger sum over all positive seven-tuples.

Since $r, s, t \geq e$, we have $rst \geq e^3$, and hence

$$\frac{1}{rst[r, s, t]} \leq e^{-5/2} \frac{(rst)^{5/6}}{rst[r, s, t]}.$$

Using (6.1), the factor after $e^{-5/2}$ is

$$\frac{1}{h^{3/2}(uvw)^{4/3}(RST)^{7/6}}.$$

Therefore

$$\begin{aligned} T_3(e) &\leq e^{-5/2} \sum_{h, u, v, w, R, S, T \geq 1} \frac{1}{h^{3/2}(uvw)^{4/3}(RST)^{7/6}} \\ &= e^{-5/2} \zeta(3/2) \zeta(4/3)^3 \zeta(7/6)^3. \end{aligned}$$

This proves the first assertion. The second follows from

$$\sum_{e > E} e^{-5/2} \leq \int_E^\infty x^{-5/2} dx = \frac{2}{3} E^{-3/2}.$$

□

For fixed $A \geq 1$ and $E \geq 1$, let $\mathcal{H}_{A,E}(X)$ be the set of integers $N \leq X$ for which there exist $e > E$ and $d \geq 1$ such that

$$N = F_e(d) \quad \text{and} \quad ed \leq AN.$$

Proposition 6.3 (Large-cofactor bound). *There is an absolute constant $C_5 > 0$ such that*

$$\#\mathcal{H}_{A,E}(X) \leq C_5 A^4 E^{-3/2} X$$

for all $A \geq 1$, $E \geq 1$, and $X \geq 1$.

Proof. If $N \in \mathcal{H}_{A,E}(X)$, choose one witnessing pair (e, d) and put $m = ed$. Lemma 3.1 gives

$$N = mg_e(m).$$

The inequality $m \leq AN$ implies $g_e(m) \geq 1/A$, and $N \leq X$ implies $m \leq AX$. Therefore, after dropping the condition $e \mid m$ and counting pairs rather than distinct values of N ,

$$\begin{aligned} \#\mathcal{H}_{A,E}(X) &\leq \sum_{m \leq AX} \sum_{e > E} \mathbf{1}_{\{g_e(m) \geq 1/A\}} \\ &\leq A^3 \sum_{m \leq AX} \sum_{e > E} g_e(m)^3 \\ &\leq A^3(AX) \sum_{e > E} T_3(e) \\ &\leq C_5 A^4 E^{-3/2} X, \end{aligned}$$

by Lemmas 6.1 and 6.2. □

7 Almost every odd integer is represented

The construction below is the two-prime instance of the prime-product divisor-prefix gadget recorded in the OEIS entry and used in jif's representability argument [13, 17, 18]. Only the density-one binary Goldbach theorem is required here.

Lemma 7.1. *All but $o(X)$ odd integers $N \leq X$ belong to $\mathcal{R}$.*

Proof. By (A3), all but $o(X)$ even integers $M \leq X$ are sums of two primes. The even integers of the form $2p$ with p prime are $o(X)$ in number because $\pi(X) = o(X)$. Therefore all but $o(X)$ even integers $M \leq X$ can be written

$$M = p + q$$

with distinct primes $p < q$.

For every such M , put $N = M + 1$ and $m = pq$. The divisors of m begin

$$1, p, q, pq,$$

so

$$N = 1 + p + q$$

is a divisor-prefix sum of m . Translating from M to $N = M + 1$ proves the claim. □

8 Proof of the main theorem

Proof of Theorem 1.1. Fix $A \geq 1$. Let C_5 be the constant in Proposition 6.3. By Lemma 5.4,

$$\delta_E \geq \frac{c_1}{E \log(2E)}.$$

Hence

$$\frac{C_5 A^4 E^{-3/2}}{\delta_E} \leq \frac{C_5}{c_1} A^4 \frac{\log(2E)}{\sqrt{E}} \longrightarrow 0 \quad (E \rightarrow \infty).$$

Choose and fix an integer $E = E(A) \geq 2$ sufficiently large that

$$C_5 A^4 E^{-3/2} \leq \frac{1}{4} \delta_E. \tag{8.1}$$

Let

$$\mathcal{S}_E(X) := \{N \leq X : (N, Q_E) = 1\}.$$

By Lemma 5.4,

$$\#\mathcal{S}_E(X) = \delta_E X + O_E(1). \quad (8.2)$$

Let $\mathcal{L}_E(X)$ be the subset of $\mathcal{S}_E(X)$ consisting of integers having a representation $N = F_e(d)$ with $e \leq E$. Proposition 5.3 gives

$$\#\mathcal{L}_E(X) = o_E(X). \quad (8.3)$$

By Proposition 6.3 and (8.1),

$$\#\mathcal{H}_{A,E}(X) \leq \frac{1}{4}\delta_E X. \quad (8.4)$$

Finally, because $2 \mid Q_E$, every integer in $\mathcal{S}_E(X)$ is odd. By Lemma 7.1, the number of integers in $\mathcal{S}_E(X)$ that are not known from that lemma to be represented is $o(X)$.

Combining (8.2)–(8.4), we find that, for all sufficiently large X , at least

$$\frac{1}{2}\delta_E X$$

integers $N \leq X$ satisfy all of the following:

- (i) $(N, Q_E) = 1$;
- (ii) $N \in \mathcal{R}$;
- (iii) N has no representation with cofactor $e \leq E$;
- (iv) N has no representation $N = F_e(d)$ with $e > E$ and $ed \leq AN$.

For any such N , suppose for contradiction that $f(N) \leq AN$. Choose the endpoint d of a prefix representation attaining $m = f(N)$ and put $e = m/d$. By Lemma 3.1, $N = F_e(d)$. If $e \leq E$, this contradicts (iii); if $e > E$, then $ed = m \leq AN$, contradicting (iv). Therefore $f(N) > AN$.

It follows that

$$\liminf_{X \rightarrow \infty} \frac{1}{X} \#\{N \leq X : N \in \mathcal{R}, f(N) > AN\} \geq \frac{1}{2}\delta_E > 0.$$

This proves the first assertion with $c_A = \delta_{E(A)}/2$.

Since the conclusion holds for every positive integer A , for each A there are infinitely many represented N with $f(N)/N > A$. Hence the limsup over represented N is infinite. $\square$

Corollary 8.1 (Shifted Erdős–Graham convention). *Let $f_0(N)$ be the least m for which N is a sum of an initial segment of the divisors of m that are greater than 1. Then*

$$\limsup_{\substack{N \rightarrow \infty \\ f_0(N) < \infty}} \frac{f_0(N)}{N} = \infty.$$

Proof. For $N \geq 2$, including the divisor 1 adds exactly 1 to every nonempty prefix sum, so

$$f(N) = f_0(N - 1)$$

whenever either side is finite. The result follows from Theorem 1.1, since replacing N by $N - 1$ does not affect unboundedness of the ratio. $\square$

Remark 8.2. The proof establishes more than the formal statement $\limsup f(N)/N = \infty$: for every fixed A , the set of represented integers satisfying $f(N) > AN$ has positive lower density. It also avoids interpreting $f(N) = \infty$ at unrepresented integers; the surviving integers are represented by an explicit Goldbach construction.

References

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